MODULAR AND INTEGRAL SPECTRAL ALGEBRAS OF ABELIAN CAYLEY GRAPHS

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Abstract. The algebra generated by the adjacency matrix of a Cayley graph on a finite abelian group is semisimple in characteristic p exactly when every element of the connection set has order prime to p. For cycles we find its minimal polynomial in every characteristic and the index of the integral algebra in the maximal order, and we show that a prism over an m-cycle has as many distinct eigenvalues as an abelian group of order $2m$ allows if and only if neither 4 nor 6 divides m.

1. Introduction

Let G be a finite abelian group of order N, let $S \subseteq G \setminus \{0\}$, and let $\Gamma = \mathrm{Cay}(G, S)$ be the Cayley graph with an arc from g to $g+s$ for every $g \in G$ and $s \in S$. For a commutative ring R, the *spectral algebra* $\mathcal{A}_R(\Gamma) = R[A]$ is the subalgebra of $G \times G$ matrices over R generated by the adjacency matrix A of Γ. When $S = -S$ and K is a field whose characteristic does not divide N, the structure of $\mathcal{A}_K(\Gamma)$ is given by the character-orbit decomposition of Bhattacharjee, Mandal and Bhattacharya [4, Theorem 6.1]: it is a product of fields, one for each orbit of $\mathrm{Gal}(\overline{K}/K)$ on the eigenvalues

$$\lambda_\chi = \sum_{s \in S} \chi(s), \qquad \chi \in \widehat{G}.$$

Two cases lie outside that theorem: fields whose characteristic divides N, and the integral algebra $\mathcal{A}_{\mathbb{Z}}(\Gamma) = \mathbb{Z}[A]$, which is an order in $\mathcal{A}_{\mathbb{Q}}(\Gamma)$. This paper treats both, and answers two questions about the rational algebra raised in [4].

Characteristic p. Write $G = G_{p'} \times G_p$, where G_p is the Sylow p-subgroup and $G_{p'}$ consists of the elements of order prime to p, and let $\pi\colon G \to G_{p'}$ be the projection. The *shadow* of Γ is the Cayley multigraph $\Gamma_{p'} = \mathrm{Cay}(G_{p'}, \pi(S))$, in which h is joined to $h + \pi(s)$ once for every $s \in S$.

Theorem A. *Let K be a field of characteristic $p > 0$ and let q be a power of p that is divisible by the exponent of G_p and congruent to 1 modulo the exponent of $G_{p'}$. Then $A^q = \sum_{s \in S} T_{\pi(s)}$ is the adjacency matrix of $\mathrm{Cay}(G, \pi(S))$, a disjoint union of $|G_p|$ copies of the shadow $\Gamma_{p'}$, the matrix $A - A^q$ is nilpotent, and*

$$\mathrm{J}(\mathcal{A}_K(\Gamma)) = (A - A^q)\,\mathcal{A}_K(\Gamma), \qquad \mathcal{A}_K(\Gamma)/\,\mathrm{J}(\mathcal{A}_K(\Gamma)) \cong \mathcal{A}_K(\Gamma_{p'}).$$

In particular $\mathcal{A}_K(\Gamma)$ is semisimple if and only if every element of S has order prime to p.

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Here J is the Jacobson radical. The description of the quotient is [4, Proposition 14.4]; the new points are that the radical is principal with an explicit generator and the semisimplicity criterion, which needs neither $S = -S$ nor S to generate G. A connected Cayley graph is semisimple in characteristic p exactly when $p \nmid N$, but a disconnected one can be semisimple when $p \mid N$: the perfect matching $\mathrm{Cay}(\mathbb{Z}/6, \{3\})$ is semisimple over $\mathbb{F}_3$.

For cycles $C_n = \mathrm{Cay}(\mathbb{Z}/n, \{\pm 1\})$ the radical can be described completely. Let $\Psi_d \in \mathbb{Z}[x]$ be the minimal polynomial of $2\cos(2\pi/d)$, so that $\Psi_1 = x - 2$, $\Psi_2 = x + 2$ and $\deg \Psi_d = \varphi(d)/2$ for $d \geq 3$. Over $\mathbb{Q}$ the minimal polynomial of A_{C_n} is $\prod_{d \mid n} \Psi_d$ (Lehmer [7], Watkins and Zeitlin [15]).

Theorem B. *Let $n = p^a m \geq 3$ with $a \geq 1$ and $p \nmid m$, put $q = p^a$, and let K have characteristic p. The minimal polynomial of A_{C_n} over K is*

$$\prod_{d \mid m} \overline{\Psi}_d(x)^{e_d}, \qquad e_d = \begin{cases} q, & d \geq 3, \\ \lceil q/2 \rceil, & d \leq 2, \end{cases}$$

where $\overline{\Psi}_d$ is the reduction of Ψ_d modulo p; these reductions are separable and pairwise coprime. Hence $\dim_K \mathcal{A}_K(C_n) = \lfloor n/2 \rfloor + 1$ if p is odd and $\dim_K \mathcal{A}_K(C_n) = n/2$ if $p = 2$.

So for odd p the minimal polynomial over $\mathbb{F}_p$ is the reduction of the rational one, while for $p = 2$ the reduction carries one superfluous factor x. Section 4 explains this through the integral algebra.

The integral algebra of a cycle. The rational algebra $\mathcal{A}_{\mathbb{Q}}(C_n) \cong \prod_{d \mid n} \mathbb{Q}(\zeta_d)^+$ has the maximal order $\mathcal{O}_n = \prod_{d \mid n} \mathcal{O}_{\mathbb{Q}(\zeta_d)^+}$, and $\mathbb{Z}[A_{C_n}]$ is a suborder of finite index

$$I(n) = [\mathcal{O}_n : \mathbb{Z}[A_{C_n}]].$$

The last display of [4] records the inclusion and leaves the index open.[1] For $d \geq 1$ put $\varepsilon_d = 1$ if $d \leq 2$ and $\varepsilon_d = 2$ if $d \geq 3$, and write $n_{p'}$ for the largest divisor of n prime to p.

Theorem C. *Let $n \geq 3$.*

(i) *$I(n) = \prod p^{2\varphi(d)/(\varepsilon_d \varepsilon_{d'})}$, the product taken over all pairs $d \mid d' \mid n$ with d'/d a power p^k, $k \geq 1$, of a prime p.*

(ii) *For a prime p with $p^a \parallel n$, $a \geq 1$,*

$$v_p(I(n)) = \begin{cases} \dfrac{n - n_{p'}}{2(p-1)} + \dfrac{a \gcd(2, n)}{2}, & p \text{ odd}, \\ \dfrac{n - n_{2'} + 2a + 1}{2}, & p = 2, \end{cases}$$

and $p \nmid I(n)$ when $p \nmid n$. In particular $\mathbb{Z}[A_{C_n}]$ is never the maximal order.

(iii) *$\mathbb{Z}[A_{C_n}]$ has index $\gcd(2, n)$ in the ring of symmetric integer circulant matrices, which is $\mathcal{A}_{\mathbb{Q}}(C_n) \cap M_n(\mathbb{Z})$.*

For instance $I(30) = 2^9 \cdot 3^6 \cdot 5^4$, read off the edges of the divisor cube of 30 in Figure 1. The proof combines a resultant identity for the polynomials Ψ_d, which for $d, d' \geq 3$ is due to Loper and Werner [9], with Apostol's evaluation of resultants of cyclotomic polynomials [1].

[1] It states that $\mathbb{Z}[A_{C_n}] = \mathcal{O}_n$ "in many (but not all) cases". Theorem C(ii) shows that equality never holds for $n \geq 3$. The examples $n = 3, 4, 5$ give $I(n) = 3, 16, 5$.

The dimension of the rational algebra. Since $\lambda_\chi = \lambda_{\bar\chi}$, the dimension of $\mathcal{A}_\mathbb{Q}(\Gamma)$, which is the number of distinct eigenvalues, is at most the number of orbits of $\chi \mapsto \bar\chi$ on $\widehat{G}$:

$$\dim_\mathbb{Q} \mathcal{A}_\mathbb{Q}(\Gamma) \le \frac{N+t}{2}, \qquad t = \#\{g \in G : 2g = 0\}. \tag{1.1}$$

This is [4, Remark 21.4], which leaves open whether noncyclic groups attain it; cycles do [4, Theorem 21.3]. On the other hand [4, Remark 19.4] asserts the cyclic value $\lfloor N/2 \rfloor + 1$ as an upper bound for every group. The prism $C_m \,\square\, K_2 = \mathrm{Cay}(\mathbb{Z}/m \times \mathbb{Z}/2, \{(\pm 1, 0), (0, 1)\})$ settles both points.

Theorem D. *The prism $C_m \,\square\, K_2$, $m \ge 3$, attains (1.1) if and only if $4 \nmid m$ and $6 \nmid m$.*

For $m \equiv 2 \pmod 4$ with $3 \nmid m$ the group $\mathbb{Z}/m \times \mathbb{Z}/2$ is not cyclic, the bound is attained, and the dimension $m+2$ exceeds $\lfloor 2m/2 \rfloor + 1 = m+1$. The smallest case is the decagonal prism with 12 distinct eigenvalues (Figure 3). In the other direction, elementary abelian 2- and 3-groups of rank at least two never attain (1.1) (Proposition 5.2).

Section 2 fixes notation. Sections 3, 4 and 5 prove Theorems A and B, Theorem C, and Theorem D. Section 6 records two facts about isomorphism types that correct statements in [4].

Related work: Mönius determined the algebraic degree [10] and the splitting field [11] of the spectrum of a circulant graph, and Babai [3] described the spectra of Cayley graphs through characters. We use [6] for background on graph spectra and [14] for cyclotomic fields.

2. Preliminaries

2.1. **The group algebra.** For a commutative ring R let $R[G]$ be the group algebra with basis $\{e_g : g \in G\}$ and $e_g e_h = e_{g+h}$. The regular representation $e_g \mapsto T_g$, where T_g is the permutation matrix of $x \mapsto x + g$, is an injective algebra homomorphism $R[G] \to M_G(R)$, and $A = \sum_{s \in S} T_s$. Hence $\mathcal{A}_R(\Gamma) \cong R[a]$ with

$$a = \sum_{s \in S} e_s \in R[G],$$

and the minimal polynomial of A over a field is that of a in the group algebra. For a multiset S the same formula defines the adjacency matrix of the Cayley multigraph; this is how the shadow in Theorem A is to be read.

2.2. **Cyclotomic notation.** For $d \ge 1$ let $\zeta_d = e^{2\pi i/d}$, let Φ_d be the d-th cyclotomic polynomial, and let Ψ_d be the minimal polynomial of $\zeta_d + \zeta_d^{-1} = 2\cos(2\pi/d)$ over $\mathbb{Q}$. Then Ψ_d is monic with integer coefficients [7], of degree $\varphi(d)/\varepsilon_d$, and

$$\Psi_d(x)^{\varepsilon_d} = \prod_{\substack{1 \le j \le d \\ \gcd(j,d)=1}} (x - \zeta_d^j - \zeta_d^{-j}), \tag{2.1}$$

since j and $d-j$ give the same root and are distinct exactly when $d \ge 3$. The ring of integers of $\mathbb{Q}(\zeta_d)^+ = \mathbb{Q}(\zeta_d + \zeta_d^{-1})$ is $\mathbb{Z}[\zeta_d + \zeta_d^{-1}] \cong \mathbb{Z}[x]/(\Psi_d)$; see [14, Proposition 2.16] and [8]. The eigenvalues of C_n are $2\cos(2\pi k/n)$, $0 \le k < n$, so

$$m_{C_n}(x) = \prod_{d \mid n} \Psi_d(x), \qquad \mathcal{A}_\mathbb{Q}(C_n) \cong \prod_{d \mid n} \mathbb{Q}(\zeta_d)^+, \qquad \dim_\mathbb{Q} \mathcal{A}_\mathbb{Q}(C_n) = \lfloor n/2 \rfloor + 1. \tag{2.2}$$

3. Characteristic p

Throughout this section K has characteristic $p > 0$, $H = G_{p'}$, $P = G_p$, and q is as in Theorem A. Such q exist: if p^b is the exponent of P and r is the order of p modulo the exponent of H, any p^k with $r \mid k$ and $k \geq b$ will do.

Proof of Theorem A. Every element of G is $g = \pi(g) + \rho(g)$ with $\rho(g) \in P$, and $qg = \pi(g)$ because $q\rho(g) = 0$ and $q\pi(g) = \pi(g)$. Since $K[G]$ is commutative of characteristic p, the q-th power map is additive, and the coefficients of a lie in the prime field, so

$$a^q = \sum_{s \in S} e_{qs} = \sum_{s \in S} e_{\pi(s)} =: a_s.$$

This element of $K[H] \subseteq K[G]$ is the one attached to the shadow, and it lies in $K[a]$.

Put $a_n = a - a_s = \sum_{s \in S} e_{\pi(s)}(e_{\rho(s)} - 1)$. The augmentation ideal of $K[P]$ is generated by the elements $e_x - 1$, $x \in P$, which satisfy $(e_x - 1)^{p^b} = e_{p^b x} - 1 = 0$. Since $K[G]$ is commutative, the ideal of $K[G]$ generated by these elements is nilpotent, and a_n lies in it. Expanding polynomials in $a = a_s + a_n$ gives

$$K[a] = K[a_s] + a_n K[a].$$

The ideal $a_n K[a]$ is nilpotent, hence contained in $\mathrm{J}(K[a])$. On the other hand $K[a_s]$ is a subalgebra of $K[H]$, which is a product of fields after extending scalars because $p \nmid |H|$ (Maschke's theorem); so $K[a_s]$ has no nonzero nilpotent elements. Its intersection with $a_n K[a]$ consists of nilpotent elements and is therefore zero, so $K[a]/a_n K[a] \cong K[a_s]$. A finite-dimensional commutative algebra without nilpotents is semisimple, which gives $\mathrm{J}(K[a]) = a_n K[a]$ and the description of the quotient.

Finally $K[a]$ is semisimple if and only if $a_n = 0$, that is, $\sum_{s \in S} e_s = \sum_{s \in S} e_{\pi(s)}$. The right side is supported on H. If some $s_0 \in S$ lies outside H, the coefficient of e_{s_0} is 1 on the left and 0 on the right. If $S \subseteq H$, then $\pi(s) = s$ for every s and the two sides agree. □

Remark 3.1. In matrix language $A = A^q + (A - A^q)$ is the Jordan–Chevalley decomposition of A, with both parts polynomials in A over the prime field. The proof also gives $\dim_K \mathrm{J}(\mathcal{A}_K(\Gamma)) = \deg m_{\Gamma,K} - \deg m_{\Gamma_{p'},K}$, where $m_{\Gamma,K}$ is the minimal polynomial over K. When S generates G, the set $S \subseteq H$ forces $H = G$, so a connected Cayley graph is semisimple in characteristic p exactly when $p \nmid N$; for cycles this is also observed in [5].

Example 3.2. Over $\mathbb{F}_3$, the graph $\mathrm{Cay}(\mathbb{Z}/6, \{3\})$ has $A^2 = I$ and minimal polynomial $(x - 1)(x + 1)$, which is separable: 3 has order 2. The graph $\mathrm{Cay}(\mathbb{Z}/6, \{2, 4\})$, two disjoint triangles, has $A^2 = A + 2I$, so $m(x) = x^2 - x - 2 = (x + 1)^2$ over $\mathbb{F}_3$, and $q = 3$ gives $A^3 = 2I$, the adjacency matrix of the shadow $\mathrm{Cay}(\mathbb{Z}/2, \{0, 0\})$, which has two loops at each vertex, with $\mathrm{J} = (A - A^3)\mathbb{F}_3[A] = (A + I)\mathbb{F}_3[A]$.

Proof of Theorem B. The minimal polynomial does not change under field extension, so we may assume K algebraically closed. Write $\mathbb{Z}/n = H \times P$ with $H \cong \mathbb{Z}/m$ and $P \cong \mathbb{Z}/q$, and let $g = (h_0, x_0)$ be the image of 1. Put $y = e_{x_0} - 1$; then $K[P] = K[y]/(y^q)$, because $y^q = e_{qx_0} - 1 = 0$ and $K[P]$ has dimension q. The characters ψ of H with values in K correspond bijectively to $\zeta = \psi(h_0)$ in the group $\mu_m(K)$ of m-th roots of unity, which has m elements because $p \nmid m$, and

$$K[G] = K[H] \otimes K[P] \xrightarrow{\sim} \prod_{\psi \in \widehat{H}} K[P], \qquad e_h \otimes u \mapsto (\psi(h)u)_\psi.$$

The image of $a = e_g + e_{-g}$ in the factor of ψ is $a_\psi = \zeta(1+y) + \zeta^{-1}(1+y)^{-1}$, and with $\lambda_\zeta = \zeta + \zeta^{-1}$ a short computation gives

$$a_\psi - \lambda_\zeta = y(1+y)^{-1}\big((\zeta - \zeta^{-1}) + \zeta y\big).$$

If $\zeta^2 \neq 1$ the bracket is a unit, so $a_\psi - \lambda_\zeta$ is y times a unit and has nilpotency index q. If $\zeta = \pm 1$ then $a_\psi - \lambda_\zeta = \zeta y^2(1+y)^{-1}$ has nilpotency index $\lceil q/2 \rceil$. The minimal polynomial of a_ψ is therefore $(x - \lambda_\zeta)^{\nu_\zeta}$ with $\nu_\zeta \in \{q, \lceil q/2 \rceil\}$ as just described, and the minimal polynomial of a is the least common multiple of these over $\zeta \in \mu_m(K)$. Now $\lambda_\zeta = \lambda_\xi$ if and only if $\xi \in \{\zeta, \zeta^{-1}\}$, since ζ and ζ^{-1} are the roots of $x^2 - \lambda_\zeta x + 1$. So the minimal polynomial is $\prod (x - \lambda_\zeta)^{\nu_\zeta}$ over the pairs $\{\zeta, \zeta^{-1}\}$.

Group the pairs by the order $d \mid m$ of ζ. In $\mathbb{Z}[\zeta_d][x]$ the polynomial $\Psi_d(x)$ is the product of $x - \omega - \omega^{-1}$ over the pairs $\{\omega, \omega^{-1}\}$ of primitive d-th roots of unity, and reduction modulo a prime above p maps these roots bijectively onto the primitive d-th roots of unity in K, because $x^d - 1$ is separable modulo p. So the product of $x - \lambda_\zeta$ over the pairs of order d is $\overline{\Psi}_d$. Its roots are distinct, and different d give coprime factors, because different pairs give different λ_ζ. Since $\zeta^2 = 1$ exactly when $d \leq 2$, the exponents are e_d.

For the dimension let $t = \gcd(2, m)$ be the number of divisors $d \leq 2$ of m. The divisors $d \geq 3$ contribute $\varphi(d)/2$ each, $(m - t)/2$ in all, so the degree of the minimal polynomial is

$$t\lceil q/2 \rceil + q(m - t)/2.$$

For odd p this is $(n + t)/2 = \lfloor n/2 \rfloor + 1$, because $t = \gcd(2, n)$. For $p = 2$, m is odd, $t = 1$ and the degree is $q/2 + q(m - 1)/2 = n/2$. □

Corollary 3.3. *For odd p the minimal polynomial of A_{C_n} over $\mathbb{F}_p$ is $\prod_{d|n} \Psi_d$ reduced modulo p. For $p = 2$ and even n it is that reduction divided by x.*

Proof. The reduction $\overline{m}$ of m_{C_n} annihilates A, so the minimal polynomial over $\mathbb{F}_p$ divides it. For odd p both are monic of degree $\lfloor n/2 \rfloor + 1$. For $p = 2$ the quotient has degree one, so it suffices to compare the multiplicities of the root 0. A root $\zeta + \zeta^{-1}$ of Ψ_d reduces to 0 modulo a prime above 2 exactly when ζ reduces to 1, that is, when d is a power of 2; so the multiplicity in $\overline{m}$ is $\sum_{b=0}^{a} \deg \Psi_{2^b} = 2^{a-1} + 1$, against 2^{a-1} in the minimal polynomial. □

4. The integral algebra of a cycle

4.1. **Symmetric circulants.** For $0 \leq k \leq n/2$ let $c_k = T_k + T_{-k}$ if $0 < k < n/2$, and $c_0 = I$, $c_{n/2} = T_{n/2}$ when these are defined. The matrices c_k have disjoint supports, and their $\mathbb{Z}$-span $\mathbb{Z}[C_n]^+$ is the ring of symmetric integer circulant matrices.

Proposition 4.1. *$\mathcal{A}_{\mathbb{Q}}(C_n)$ is the $\mathbb{Q}$-span of $c_0, \ldots, c_{\lfloor n/2 \rfloor}$, so $\mathbb{Z}[C_n]^+ = \mathcal{A}_{\mathbb{Q}}(C_n) \cap M_n(\mathbb{Z})$. If n is odd then $\mathbb{Z}[A] = \mathbb{Z}[C_n]^+$. If n is even then $\mathbb{Z}[A]$ is spanned by $c_0, \ldots, c_{n/2-1}$ and $2c_{n/2}$, and has index 2 in $\mathbb{Z}[C_n]^+$.*

Proof. With $A = c_1$ we have

$$Ac_k = c_{k+1} + c_{k-1} \quad (2 \leq k,\ k+1 < n/2), \qquad Ac_1 = c_2 + 2c_0 \quad (2 < n/2),$$

and, for even n, $Ac_{n/2-1} = 2c_{n/2} + c_{n/2-2}$ if $n \geq 6$ and $Ac_1 = 2c_2 + 2c_0$ if $n = 4$. By induction $A^j \in c_j + \mathbb{Z}c_0 + \cdots + \mathbb{Z}c_{j-1}$ for $j < n/2$, and $A^{n/2} \in 2c_{n/2} + \mathbb{Z}c_0 + \cdots + \mathbb{Z}c_{n/2-1}$ for even n. Because m_{C_n} is monic of degree $\lfloor n/2 \rfloor + 1$, the powers $A^0, \ldots, A^{\lfloor n/2 \rfloor}$ form a $\mathbb{Z}$-basis of $\mathbb{Z}[A]$, and the triangular relations give both statements about $\mathbb{Z}[A]$. Tensoring with $\mathbb{Q}$ gives the first sentence. □

This explains Corollary 3.3 from the integral side: the reduction map $\mathbb{Z}[A] \otimes \mathbb{F}_p \to M_n(\mathbb{F}_p)$ is injective exactly when $\mathbb{Z}[A]$ is saturated at p, which by Proposition 4.1 fails only for $p = 2$ and even n, and then the kernel is spanned by the image of $2c_{n/2}$.

4.2. **Resultants.** The next two lemmas are standard in substance; we include short proofs.

Lemma 4.2. *Let $f_1, \dots, f_r \in \mathbb{Z}[x]$ be monic and pairwise coprime in $\mathbb{Q}[x]$, and $f = f_1 \cdots f_r$. The natural map $\mathbb{Z}[x]/(f) \to \prod_i \mathbb{Z}[x]/(f_i)$ is injective with image of index $\prod_{i<j} |\operatorname{Res}(f_i, f_j)|$.*

Proof. Take $r = 2$, $f_1 = u$, $f_2 = v$. If $h \in \mathbb{Z}[x]$ is divisible by u and by v in $\mathbb{Z}[x]$, then $uv \mid h$ in $\mathbb{Q}[x]$, and division by the monic polynomial uv shows that the quotient has integer coefficients; so the map is injective. The map $(\alpha, \beta) \mapsto \alpha - \beta$ from $\mathbb{Z}[x]/(u) \times \mathbb{Z}[x]/(v)$ onto $\mathbb{Z}[x]/(u, v)$ has kernel exactly the image, because $\alpha - \beta = u\gamma + v\delta$ means that $\alpha - u\gamma = \beta + v\delta$ is a common lift. Finally $\mathbb{Z}[x]/(u, v) = R/vR$ with $R = \mathbb{Z}[x]/(u)$ free of rank $\deg u$, and $|R/vR|$ is the absolute value of the determinant of multiplication by v on R, which is $\prod_{u(\theta)=0} v(\theta) = \pm \operatorname{Res}(u, v) \neq 0$. For $r > 2$ factor the map through $\mathbb{Z}[x]/(f_1 \cdots f_{r-1}) \times \mathbb{Z}[x]/(f_r)$ and use $\operatorname{Res}(f_1 \cdots f_{r-1}, f_r) = \prod_{i<r} \operatorname{Res}(f_i, f_r)$. □

Lemma 4.3. *For $d \neq d'$,*

$$|\operatorname{Res}(\Psi_d, \Psi_{d'})|^{\varepsilon_d \varepsilon_{d'}} = \operatorname{Res}(\Phi_d, \Phi_{d'})^2.$$

Consequently, for $d < d'$, $|\operatorname{Res}(\Psi_d, \Psi_{d'})| = p^{2\varphi(d)/(\varepsilon_d \varepsilon_{d'})}$ if $d'/d = p^k$ with p prime and $k \geq 1$, and $|\operatorname{Res}(\Psi_d, \Psi_{d'})| = 1$ otherwise.

Proof. By (2.1), $\operatorname{Res}(\Psi_d, \Psi_{d'})^{\varepsilon_d \varepsilon_{d'}}$ is the product of $\alpha + \alpha^{-1} - \beta - \beta^{-1}$ over all primitive d-th roots α and primitive d'-th roots β. The identity

$$\alpha + \alpha^{-1} - \beta - \beta^{-1} = (\alpha - \beta)(1 - \alpha^{-1}\beta^{-1}) = -\beta^{-1}(\alpha - \beta)(\alpha^{-1} - \beta)$$

and the fact that $\alpha \mapsto \alpha^{-1}$ permutes the primitive d-th roots show that this product is a root of unity times $\operatorname{Res}(\Phi_d, \Phi_{d'})^2$. Taking absolute values gives the identity. Apostol [1] proved that for $1 \leq d < d'$ the resultant $\operatorname{Res}(\Phi_d, \Phi_{d'})$ equals $p^{\varphi(d)}$ if d'/d is a power of a prime p and 1 otherwise.[2] The exponent $2\varphi(d)/(\varepsilon_d \varepsilon_{d'})$ is an integer because $\varphi(d)$ is even for $d \geq 3$. □

For $d, d' \geq 3$ the second statement is [9, Theorem 3.4], where the sign of the resultant is also determined.

4.3. **Proof of Theorem C.** By (2.2) and Section 2 we have $\mathbb{Z}[A] \cong \mathbb{Z}[x]/(m_{C_n})$ and

$$\mathcal{O}_n \cong \prod_{d \mid n} \mathbb{Z}[x]/(\Psi_d),$$

compatibly with the inclusion. Lemmas 4.2 and 4.3 give (i).

For (ii) fix p with $p^a \parallel n$, $a \geq 1$, write $m = n_{p'}$, and collect the pairs in (i) with d'/d a power of p: they are $d = p^b e$, $d' = p^c e$ with $e \mid m$ and $0 \leq b < c \leq a$. If $e \geq 3$ every ε equals 2 and the contribution is

$$\frac{\varphi(e)}{2} \sum_{0 \leq b < c \leq a} \varphi(p^b) = \frac{\varphi(e)}{2} \sum_{c=1}^{a} p^{c-1} = \frac{\varphi(e)}{2} \cdot \frac{p^a - 1}{p - 1}.$$

[2] For $d = 1$ this is the classical value $\Phi_{d'}(1)$.

Summing over $e \mid m$, $e \geq 3$, gives $\frac{m-t}{2} \cdot \frac{p^a-1}{p-1}$ with $t = \gcd(2,m)$. If $e \leq 2$ and p is odd, the pairs with $b = 0$ contribute 1 each, a in all, and those with $b \geq 1$ contribute $\varphi(p^b)/2$, in all $\frac{1}{2}\sum_{c=1}^{a}(p^{c-1}-1)$; the total for one such e is $\frac{1}{2}\big(a + \frac{p^a-1}{p-1}\big)$. Adding up,

$$v_p(I(n)) = \frac{m-t}{2} \cdot \frac{p^a-1}{p-1} + \frac{t}{2}\Big(a + \frac{p^a-1}{p-1}\Big) = \frac{(p^a-1)\,m}{2(p-1)} + \frac{at}{2},$$

which is the stated formula since $(p^a-1)m = n - n_{p'}$ and $t = \gcd(2,n)$ for odd p. For $p = 2$ only $e = 1$ has $e \leq 2$; its pairs contribute 2 for $(b,c) = (0,1)$, 1 for each $(0,c)$ and $(1,c)$ with $c \geq 2$, and 2^{b-2} for each (b,c) with $b \geq 2$, in all $2 + 2(a-1) + \sum_{c=3}^{a}(2^{c-2}-1) = a + 2^{a-1}$. Together with $(2^a-1)(m-1)/2$ from the odd $e \geq 3$ this is $(n-m+2a+1)/2$. Both expressions are positive, so $I(n) > 1$ whenever $n \geq 3$; if $p \nmid n$ no pair in (i) involves p.

Part (iii) is Proposition 4.1. □

Corollary 4.4. *The discriminant of m_{C_n} is $I(n)^2 \prod_{d\mid n} \operatorname{disc}(\mathbb{Q}(\zeta_d)^+)$, the order $\mathbb{Z}[C_n]^+$ has index $I(n)/\gcd(2,n)$ in $\mathcal{O}_n$, and*

$$\log I(n) = \frac{n}{2}\sum_{p\mid n}\Big(1 - \frac{n_{p'}}{n}\Big)\frac{\log p}{p-1} + O(\log n).$$

Proof. The first statement is the usual relation between discriminant and index [13, Chapter I, (2.12)], applied factor by factor; the second follows from Theorem C(iii); the third from (ii), since the terms $\frac{a}{2}\gcd(2,n)\log p$ and $\frac{2a+1}{2}\log 2$ add up to at most $\log n + 1$. □

n	$I(n)$	n	$I(n)$	n	$I(n)$
3	3	13	13	23	23
4	2^4	14	$2^5\,7^2$	24	$2^{14}\,3^5$
5	5	15	$3^3\,5^2$	25	5^4
6	$2^3\,3^2$	16	2^{12}	26	$2^8\,13^2$
7	7	17	17	27	3^8
8	2^7	18	$2^6\,3^6$	28	$2^{13}\,7^3$
9	3^3	19	19	29	29
10	$2^4\,5^2$	20	$2^{10}\,5^3$	30	$2^9\,3^6\,5^4$
11	11	21	$3^4\,7^2$		
12	$2^7\,3^3$	22	$2^7\,11^2$		

TABLE 1. The index $I(n) = [\mathcal{O}_n : \mathbb{Z}[A_{C_n}]]$ for $3 \leq n \leq 30$. Each value agrees with $\sqrt{|\operatorname{disc}(m_{C_n})|/\prod_{d\mid n}|\operatorname{disc}\mathbb{Q}(\zeta_d)^+|}$, computed independently for $n \leq 150$.

5. The dimension of the rational algebra

Recall (1.1): $\widehat{G} \cong G$, and the characters fixed by $\chi \mapsto \bar{\chi}$ are those of order at most two, of which there are t; so the involution has $(N+t)/2$ orbits.

Lemma 5.1. *Let $\gamma = 2\cos(2\pi j/d)$ with $\gcd(j,d) = 1$. Every Galois conjugate of γ is ≥ 0 if and only if $d \in \{1,4,6\}$, and every conjugate is ≤ 0 if and only if $d \in \{2,3,4\}$.*

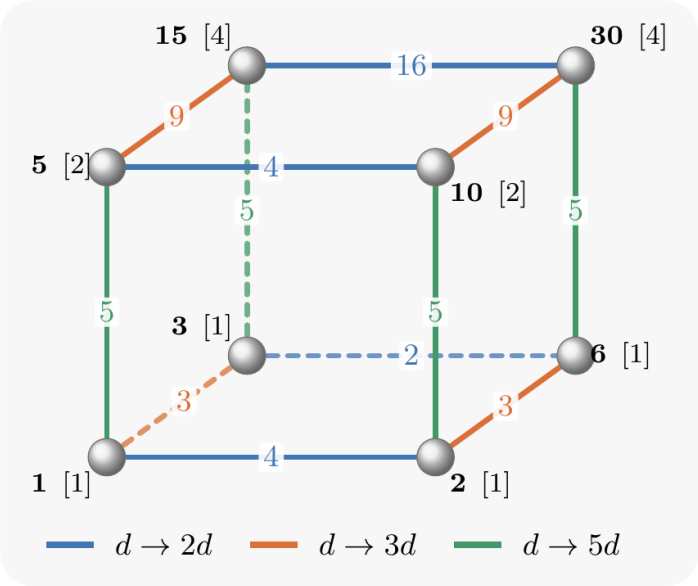


FIGURE 1. The divisors of 30 as a cube. The edge from d to pd carries $|\operatorname{Res}(\Psi_d, \Psi_{pd})|$, and the product of all twelve labels is $I(30) = 2^9 \cdot 3^6 \cdot 5^4$. Vertices are labelled by d and the degree of $\mathbb{Q}(\zeta_d)^+$.

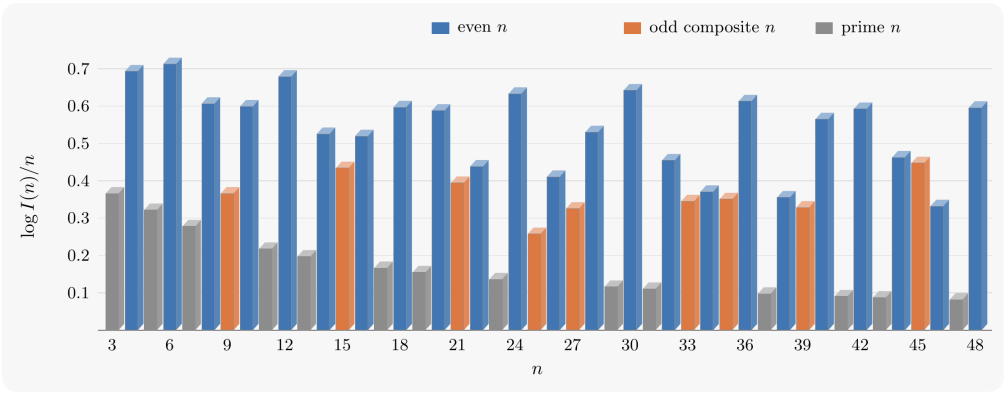


FIGURE 2. The normalized index $\log I(n)/n$ for $3 \le n \le 48$. For prime n the value is $(\log n)/n$; the largest values occur for even n with several small prime factors.

Proof. The conjugates are $2\cos(2\pi i/d)$ with $\gcd(i,d) = 1$, and such a value is negative exactly when $d/4 < i < 3d/4$. For odd $d \ge 3$ take $i = (d-1)/2$; for $d = 2e$ with $e \ge 4$ even take $i = e-1$, and with $e \ge 5$ odd take $i = e-2$. In each case $\gcd(i,d) = 1$ and $d/4 < i < 3d/4$, so some conjugate is negative. The remaining d are $1, 2, 4, 6$, with conjugates $\{2\}$, $\{-2\}$, $\{0\}$, $\{1\}$. For the second statement, $\cos(2\pi/d) \le 0$ forces $d \le 4$, and among $d = 2, 3, 4$ the conjugates are $\{-2\}$, $\{-1\}$, $\{0\}$, while $d = 1$ gives 2. □

Proof of Theorem D. Here $N = 2m$, and $t = 4$ for even m and $t = 2$ for odd m, so $(N+t)/2 = 2(\lfloor m/2 \rfloor + 1)$. The eigenvalues of the prism are $\gamma_k \pm 1$ with $\gamma_k = 2\cos(2\pi k/m)$, $0 \le k \le \lfloor m/2 \rfloor$,

and the γ_k are distinct. So the bound is attained if and only if $\gamma_k + 1 \neq \gamma_j - 1$ for all j, k, that is, no two of the γ's differ by 2. If $4 \mid m$ then $\gamma_0 - \gamma_{m/4} = 2 - 0$, and if $6 \mid m$ then $\gamma_{m/6} - \gamma_{m/3} = 1 - (-1)$. Conversely suppose $\gamma_j - \gamma_k = 2$. For every automorphism σ of $\mathbb{Q}(\zeta_m)$ we get $\sigma(\gamma_j) - \sigma(\gamma_k) = 2$, and since every conjugate lies in $[-2, 2]$ this forces $\sigma(\gamma_j) \geq 0 \geq \sigma(\gamma_k)$. By Lemma 5.1, $\gamma_j \in \{2, 0, 1\}$ and $\gamma_k \in \{-2, 0, -1\}$, so (γ_j, γ_k) is $(2, 0)$, $(0, -2)$ or $(1, -1)$. The first two need an element of order 4 in $\mathbb{Z}/m$, the third one of order 6. □

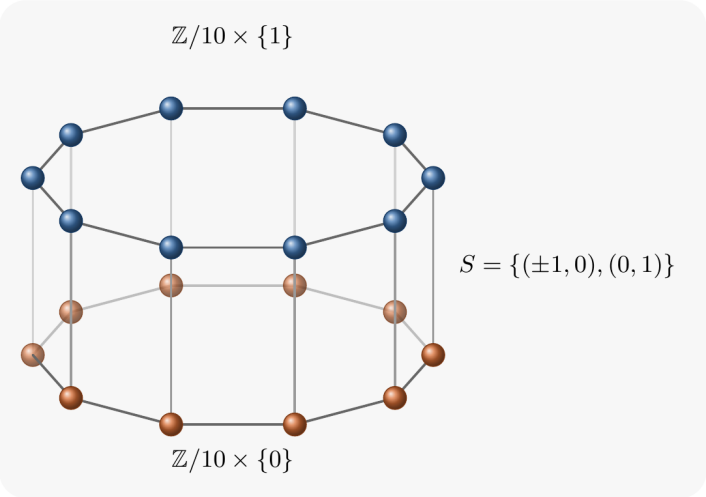


FIGURE 3. The decagonal prism $C_{10} \,\square\, K_2$ on $\mathbb{Z}/10 \times \mathbb{Z}/2$. Its twelve distinct eigenvalues ± 3, ± 1, $\pm\phi$, $\pm(\phi + 1)$, $\pm(\phi - 1)$, $\pm(\phi - 2)$, where $\phi = (1 + \sqrt{5})/2$, reach the bound $(20 + 4)/2$.

Proposition 5.2. *Let $k \geq 2$. For $G = (\mathbb{Z}/2)^k$ every Cayley graph has $\dim_{\mathbb{Q}} \mathcal{A}_{\mathbb{Q}}(\Gamma) \leq 2^{k-1} + 1 < 2^k = (N + t)/2$. For $G = (\mathbb{Z}/3)^k$ every Cayley graph with $S = -S$ has $\dim_{\mathbb{Q}} \mathcal{A}_{\mathbb{Q}}(\Gamma) \leq (3^{k-1} + 3)/2 < (3^k + 1)/2 = (N + t)/2$. Both bounds are attained for $k = 2$.*

Proof. For $G = (\mathbb{Z}/2)^k$ and $\chi \neq 1$, $\lambda_\chi = 2|S \cap \ker \chi| - |S|$, and $|S \cap \ker \chi|$ lies between 0 and $2^{k-1} - 1$; with $\lambda_1 = |S|$ there are at most $2^{k-1} + 1$ values. For $G = (\mathbb{Z}/3)^k$ write $S = T \cup (-T)$ with $|T| = \ell$. Each $s \in T$ contributes $\chi(s) + \overline{\chi(s)} \in \{2, -1\}$, so $\lambda_\chi = 3|T \cap \ker \chi| - \ell$, and for $\chi \neq 1$ the count $|T \cap \ker \chi|$ is at most $(3^{k-1} - 1)/2$, the number of pairs $\pm s$ in a hyperplane. With λ_1 this gives the bound. For $k = 2$, $S = \{(1, 0), (0, 1)\}$ in $(\mathbb{Z}/2)^2$ has the eigenvalues $2, 0, -2$, and $S = \{\pm(1, 0), \pm(0, 1)\}$ in $(\mathbb{Z}/3)^2$ has $4, 1, -2$. □

Table 2 lists the largest dimension over all symmetric S for small groups, found by exhaustive search. Apart from cyclic groups, only $\mathbb{Z}/2 \times \mathbb{Z}/10$ reaches the bound in this range, in accordance with Theorem D.

Question 5.3. For which finite abelian groups is (1.1) attained by some symmetric S?

6. Isomorphism types

Two finite products of number fields are isomorphic as $\mathbb{Q}$-algebras if and only if their factors agree up to isomorphism and multiplicity, because the factors are the localizations at the minimal primes. Hence $\mathcal{A}_{\mathbb{Q}}(\mathrm{Cay}(G, S)) \cong \mathcal{A}_{\mathbb{Q}}(\mathrm{Cay}(G, S'))$ if and only if the multisets of fields

G	N	max	$(N+t)/2$	G	N	max	$(N+t)/2$
$(\mathbb{Z}/2)^2$	4	3	4	$\mathbb{Z}/2\times\mathbb{Z}/8$	16	9	10
$\mathbb{Z}/2\times\mathbb{Z}/4$	8	4	6	$\mathbb{Z}/3\times\mathbb{Z}/6$	18	7	10
$(\mathbb{Z}/2)^3$	8	4	8	$\mathbb{Z}/2\times\mathbb{Z}/10$	20	12	12
$(\mathbb{Z}/3)^2$	9	3	5	$(\mathbb{Z}/2)^2\times\mathbb{Z}/4$	16	6	12
$\mathbb{Z}/2\times\mathbb{Z}/6$	12	7	8	$(\mathbb{Z}/2)^4$	16	6	16
$\mathbb{Z}/4\times\mathbb{Z}/4$	16	6	10	$\mathbb{Z}/2\times\mathbb{Z}/12$	24	13	14
$(\mathbb{Z}/5)^2$	25	10	13	$\mathbb{Z}/3\times\mathbb{Z}/9$	27	13	14

TABLE 2. Largest number of distinct eigenvalues of $\mathrm{Cay}(G,S)$ over all symmetric $S\subseteq G\setminus\{0\}$, against the bound (1.1).

$\mathbb{Q}(\lambda_O)$, O running over the Galois orbits of eigenvalues, agree up to isomorphism. Equal orbit sizes are not enough.[3]

Example 6.1. On $\mathbb{Z}/24$ take $S=\{\pm2\}$ and $S'=\{\pm1,\pm5\}$. The first graph is two copies of C_{12}, with algebra $\mathbb{Q}^5\times\mathbb{Q}(\sqrt{3})$. For the second, $\lambda_k=2\cos(\pi k/12)+2\cos(5\pi k/12)=4\cos(\pi k/4)\cos(\pi k/6)$, which takes the rational values $4,0,2,-2,-4$ at $k=0,2,4,8,12$ and the values $\pm\sqrt{6}$ at $k=1,7$; the algebra is $\mathbb{Q}^5\times\mathbb{Q}(\sqrt{6})$. Both have dimension 7 and orbit sizes $1,1,1,1,1,2$, but they are not isomorphic.

Proposition 6.2. *For $n\geq3$ the isomorphism type of the $\mathbb{Q}$-algebra $\mathcal{A}_{\mathbb{Q}}(C_n)$ determines n.*

Proof. The dimension $D=\lfloor n/2\rfloor+1$ leaves $n\in\{2D-2,2D-1\}$. Recall that $\mathbb{Q}(\zeta_d)^+=\mathbb{Q}$ for $d\leq4$ and $d=6$, that $\mathbb{Q}(\zeta_d)^+=\mathbb{Q}(\zeta_{d/2})^+$ for $d\equiv2\pmod4$, and that otherwise the conductor of $\mathbb{Q}(\zeta_d)^+$ is d.[4] So every factor of $\mathcal{A}_{\mathbb{Q}}(C_n)$ has conductor dividing n, and when n is odd and $n\geq5$ the factor $\mathbb{Q}(\zeta_n)^+$ has conductor n. Thus $n=2D-1\geq5$ exactly when some factor has conductor $2D-1$, since $2D-1$ does not divide $2D-2$. The case $D=2$ forces $n=3$. □

The statement is [4, Proposition 18.4], whose proof uses the dimension alone.[5]

Computations and formal verification. Every formula above was checked by computer: Theorem A, including the shadow and the dimension of the radical, in characteristics 2, 3 and 5 on twelve groups of order at most 12, with every connection set for $N\leq8$ and every symmetric one beyond (2,412 cases); Theorem B against the minimal polynomial over $\mathbb{F}_p$ for $p\leq7$ and $n\leq80$; Theorem C against the discriminants of m_{C_n} and of the fields $\mathbb{Q}(\zeta_d)^+$ for $n\leq150$, and the valuation formula for $n\leq400$; Proposition 4.1 by Smith normal form for $n\leq40$; Theorem D numerically for $m\leq300$. The finite instances are verified in Lean 4 [12] by evaluation in the kernel, using only the standard axioms: the minimal polynomials of A_{C_n} over $\mathbb{Z}$ for $n\leq30$ and over $\mathbb{F}_p$ for $p\leq7$, $p\mid n\leq30$ (each by an annihilating identity and a triangular rank certificate), the identities of Example 3.2, the two algebras of Example 6.1, the number of distinct eigenvalues of every prism $C_m\,\square\,K_2$ with $m\leq30$ (so Theorem D for

[3] This corrects the "if" direction of [4, Proposition 21.1].

[4] The field $\mathbb{Q}(\zeta_d)^+$ is fixed by $\{\pm1\}\subseteq(\mathbb{Z}/d)^\times$, and it lies in $\mathbb{Q}(\zeta_f)$ for a proper divisor f of d only if the kernel of $(\mathbb{Z}/d)^\times\to(\mathbb{Z}/f)^\times$ is contained in $\{\pm1\}$. For $f\geq3$ this kernel does not contain -1, so it must be trivial, which forces $d=2f$ with f odd; for $f\leq2$ it is all of $(\mathbb{Z}/d)^\times$, which forces $\varphi(d)\leq2$.

[5] Three further corrections to [4]. In Remark 18.3, $K_{3,3}$ has spectrum $\{3,0,0,0,0,-3\}$, not that of C_6; the pair C_6, $K_3\,\square\,K_2$ on $\mathbb{Z}/6$ illustrates the intended point, with distinct eigenvalues $\{2,1,-1,-2\}$ and $\{3,1,0,-2\}$. Remark 19.4 holds for cyclic G only, by Theorem D. In Proposition 19.3 the proof needs a character of order greater than two, which $(\mathbb{Z}/2)^k$ lacks; the statement survives there by Proposition 5.2.

$m \le 30$), the indices of Table 1 for $n \le 16$ as determinants of the embedding $\mathbb{Z}[A_{C_n}] \to \mathcal{O}_n$, the product formula of Theorem C(i) for $n \le 30$ and the valuation formula for $n \le 60$. The general theorems are proved in the text and are not formalized. Programs and the Lean development are archived in [2].

Use of generative AI. The author used Claude (Anthropic) to check the results by computer and in Lean and to help draft the text; the author verified every step and is responsible for the content.

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